AI for Scientific Research: A Comprehensive Survey on Architecture, Research Lifecycle Integration, Ethics and Security

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Abstract

Scientific literature is growing fast and research workflows are getting more complicated, so demand for AI tools that help at each stage of scholarly work is rising. Existing surveys usually cover a single task, such as literature retrieval, or describe agent architectures without tying them to the research lifecycle. We address that gap with a systematic review of 40 peer-reviewed journal papers, conference papers, and preprints published from 2019 to 2026 (see Section 2). The review asks how AI systems are designed, deployed, and evaluated across the research workflow, from topic exploration and literature review through experimentation, writing, peer review, and governance. Using a coding methodology based on PRISMA 2020, we group the literature into four themes: research guidance and lifecycle integration, agentic and retrieval-augmented architectures, tool ecosystems, and ethics and security. Hybrid modular architectures, which combine retrieval-augmented generation, multi-agent coordination, and human oversight, are more often reported to outperform monolithic or fully autonomous designs. That pattern rests on qualitative comparisons inside individual primary studies. No controlled, head-to-head benchmark compares the architectures directly. The corpus also points to four research gaps: there are no standardized end-to-end benchmarks, factual grounding in multi-step workflows is unresolved, long-horizon memory and cross-document reasoning are underdeveloped, and

explainability and auditability are limited. We then propose an architectural-lifecycle framework that maps six research-lifecycle stages against the four levels of the agent-architecture taxonomy developed in this survey. Section 1 explains how it differs from prior surveys. The review follows PRISMA 2020 [41], with the checklist and search-and-screening details in Section 2. We did not pre-register a protocol, since this is a structured academic survey and not a clinical systematic review.

Keywords: AI in Scientific Research, Automated Literature Review, Research Automation, Research Agents, Scientific Writing Assistance, Paper Summarization, Scientific Document Analysis, Scholarly Data Processing, AI Ethics in Research

1 Introduction

Artificial intelligence is moving from isolated task automation toward systems that take part in broader research workflows. In scholarly settings, the growing volume of publications, the complexity of disciplinary knowledge, and the demand for faster synthesis have made manual literature processing harder to sustain. As a result, AI now supports individual subtasks including fuller research processes, topic discovery, literature review, evidence extraction, scientific writing, and review support [1–3].

A recurring theme in the literature is AI emerging as a research companion. Systems such as ARC help students identify suitable topics and organize early-stage research activity [1]. Other systems focus on literature mining, paper screening, and structured extraction of experimental knowledge from scientific texts [2]. A third line of work automates parts of systematic literature review workflows while leaving final inclusion, interpretation, and synthesis decisions to humans [3]. Together, these studies suggest AI is becoming a working part of the research lifecycle.

This survey addresses a gap in the literature by organizing these developments into a coherent architectural and lifecycle-based framework. Prior surveys cover narrower ground: Bolaños et al. [3] focus only on literature review automation, Gridach et al. [12] examine agentic discovery without addressing governance, and Chen et al. [14] survey AI4Research tasks without an architectural taxonomy. This work spans the full research lifecycle, integrates architectural analysis with lifecycle coverage, and treats ethics and security as design constraints rather than afterthoughts. This survey is built around four research questions:

- **RQ1.** How is AI currently used across the stages of the scientific research lifecycle, from topic exploration through peer review and governance, and how mature is each application area (Section 4)?
- **RQ2.** What architectural patterns underlie research-oriented AI agents, and how do component techniques, agent patterns, operational-management layers, and deployment-scale architectures trade off against each other (Section 5)?
- **RQ3.** Within the reviewed corpus, does the evidence favor monolithic or hybrid modular agent architectures for scholarly tasks (Section 5)?

The authors declare no conflict of interest.

- **RQ4.** What gaps in evaluation, grounding, memory, and auditability limit how far research AI agents can be trusted, and what design directions follow from them (Section 9)?

In answering these questions, the survey makes five differentiated contributions:

1. **Unified lifecycle taxonomy.** We map 40 papers onto a six-stage research lifecycle (Section 4). We keep the sequential stages analytically distinct from cross-cutting support activities (e.g., funding) and research-method categories (e.g., qualitative-research tooling), and for each stage we note which AI capabilities are mature, which are emergent, and which remain unsupported. This gives more complete coverage than the single- or dual-task framing used in prior surveys.
2. **Architectural taxonomy with trade-off analysis.** We compare twelve agent architecture families, from classical reactive models through Agentic RAG, TacTool, and AgentOps, and assess their suitability for scholarly use cases. Gridach et al. and AI4Research offer descriptive inventories; we go further and weigh the trade-offs between families.
3. **Expanded tool ecosystem review.** We evaluate twelve AI research tools, against three in most prior surveys, covering both discovery-oriented platforms and synthesis/communication tools. We also assess several tools' promotional claims critically.
4. **Four operationalized research gaps.** Each gap is grounded in specific cited evidence and points to a concrete design direction. This offers actionable guidance for addressing AI limitations.
5. **An explicit architectural-lifecycle framework.** We synthesize the lifecycle taxonomy and the architectural taxonomy into a single framework (Section 6) that cross-maps lifecycle stage against architectural level, something none of Bolaños et al., Gridach et al., or Chen et al. attempt.

2 Survey Methodology

2.1 Search Strategy and Information Sources

Searches covered five electronic databases: IEEE Xplore, ACM Digital Library, arXiv (cs.AI, cs.IR, cs.CL), Google Scholar, and Semantic Scholar. The main search concepts were: *agent / agentic AI / multi-agent system*, *scientific research / literature review / scholarly workflow*, and *RAG / LLM / NLP*. These concepts were combined into five conceptual query lines, then translated into each database's own query syntax, since a single string copied verbatim across interfaces does not parse identically on all five platforms (field-tag conventions, wildcarding, and boolean-operator precedence all differ). Table 1 gives the exact, database-specific string used for one representative query line (*agentic AI research agents for literature review*); the remaining four conceptual lines were translated the same way.

No language or publication-type filters were applied beyond the eligibility criteria in Table 3. The initial search was run on 30 June 2025 for publications from 1 January 2019 onward. Because the field moves quickly, we ran a supplementary search

Table 1 Representative Database-Specific Search String (Query Line 1 of 5: "agentic AI research agents for literature review")

Database	Query string as executed
IEEE Xplore	`("Full Text & Metadata":"agentic AI" OR "agent-based") AND ("Full Text & Metadata":"literature review" OR "systematic review")`
ACM Digital Library	`[Abstract: "agentic ai"] OR [Abstract: "multi-agent"] AND [Abstract: "literature review"] AND [Publication Date: 2019 TO 2026]`
arXiv (cs.AI, cs.IR, cs.CL)	`abs:"agentic AI" AND abs:"literature review" AND (cat:cs.AI OR cat:cs.IR OR cat:cs.CL)`
Google Scholar	`"agentic AI" OR "research agent" "literature review" -patent -citation`
Semantic Scholar (API/UI)	`query="agentic AI literature review"; fieldsOfStudy= Computer Science; year=2019-2026`

pass with the same five conceptual query lines, re-translated into each database's syntax as above, on 15 August 2026 to catch recent LLM-era work, including several 2026 publications (e.g., [23]) that postdate the initial search; that is why the corpus runs 2019 through mid-2026 rather than stopping in mid-2025. Earlier foundational works (e.g., the BDI framework, Contract Net Protocol) were kept when cited as direct architectural antecedents, regardless of publication year.

Table 2 reports the number of records each database contributed before and after deduplication. Deduplication ran in two passes. First, an automated pass in Zotero matched records by DOI, or by normalized title+first-author+year where no DOI existed. Second, both reviewers manually checked all automatically flagged duplicates plus all cross-database matches with title similarity above 90

Table 2 Database-Wise Search Yields Before and After Deduplication

Database	Raw hits	After dedup.
IEEE Xplore	96	84
ACM Digital Library	41	33
arXiv (cs.AI/cs.IR/cs.CL)	88	79
Google Scholar	112	71
Semantic Scholar	74	45
Total	**411**	**312**

Google Scholar and Semantic Scholar contributed the largest share of duplicate and near-duplicate records because both index preprint, conference, and journal versions of the same work separately; IEEE Xplore and ACM Digital Library, which index a single canonical record per publication, showed the smallest reduction.

2.2 Inclusion and Exclusion Criteria

Table 3 Inclusion and Exclusion Criteria Applied During Screening

Inclusion Criteria	Exclusion Criteria
Primary focus on AI/ML applied to any stage of the scientific research lifecycle	Pure theoretical works with no system or architectural contribution
Addresses agent architecture, RAG, multi-agent systems, or LLM orchestration	Industry white-papers without empirical or architectural grounding
Peer-reviewed conference or journal paper, or arXiv preprint with verifiable provenance	Duplicate or near-duplicate publications from same authors
Written in English	Works focused solely on data-science pipelines unrelated to scholarly tasks
Published 2019–2026 (initial pass to mid-2025; supplementary pass extended coverage to mid-2026, see Section 2)	Earlier works retained only if foundational (e.g., BDI framework, Contract Net)

2.3 Screening and Selection Process

Figure 1 shows the PRISMA 2020 selection process. The five-database search identified 411 raw records (Table 2); deduplication, as described above, removed 99 duplicate or near-duplicate records, leaving 312 unique records entering screening. Title and abstract screening by two independent reviewers excluded 223 records as clearly irrelevant, leaving 89 records for full-text retrieval and eligibility assessment. Full papers were assessed against the inclusion/exclusion criteria in Table 3; 49 were excluded at this stage, yielding a final corpus of 40 papers. Reviewer disagreements were resolved by discussion and, where necessary, by a third reviewer's casting vote.

Of the 49 papers excluded after full-text review, the main reasons were: (a) criteria mismatch, where the paper addressed AI in a context unrelated to the scholarly research lifecycle (e.g., industrial automation or clinical trials without research-workflow components, $n = 31$); and (b) near-duplicate publications, where an earlier conference version was superseded by a more complete journal version already in the corpus ($n = 18$). No automation tools or machine-learning classifiers were used at any stage of screening; all decisions were made by human reviewers.

2.4 Data Collection and Coding Process

Each included paper was independently coded by two reviewers using a structured coding sheet covering: (1) research lifecycle stage(s) addressed, (2) architectural pattern(s) employed, (3) primary AI technique, (4) evaluation method, (5) ethical/security concerns raised, and (6) identified limitations. Where the two primary reviewers could not reach agreement by discussion, a third reviewer adjudicated; a fourth reviewer was consulted in the small number of cases ($n = 3$) where the third reviewer's adjudication was itself contested. Inter-rater reliability was assessed using Cohen's κ; agreement exceeded $\kappa = 0.78$ [42] across all coding dimensions before adjudication, which counts

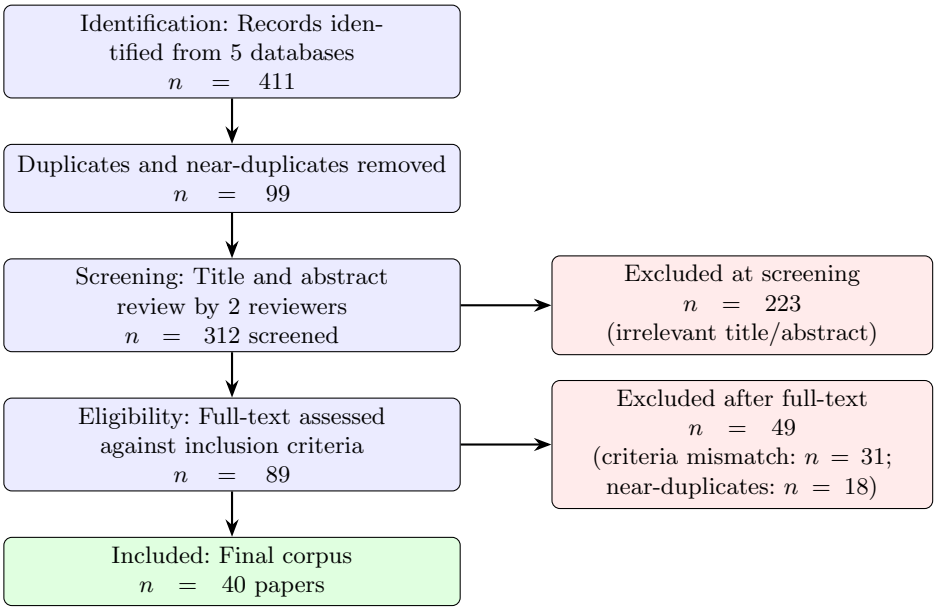


Fig. 1 PRISMA 2020 paper selection flow for the survey corpus ($n = 40$). Of 411 records identified across five databases, 99 duplicate or near-duplicate records were removed, leaving 312 unique records for title/abstract screening by two independent reviewers; disagreements were resolved by discussion or third-reviewer adjudication. Primary exclusion reasons after full-text review: criteria mismatch ($n = 31$) and near-duplicate publications ($n = 18$) that survived automated deduplication but were caught on manual full-text inspection.

as substantial agreement. Thematic categories were derived inductively from the coded data and checked against the four pillars described in the Introduction.

2.5 Quality Assessment of Included Sources

The corpus mixes source types that do not share a common peer-review or validation standard: peer-reviewed journal articles, peer-reviewed conference papers, arXiv preprints, one institutional report, and several papers whose primary contribution is a prototype system or tool description rather than an empirical study. A single risk-of-bias instrument such as the Cochrane tool or ROBIS would not fit this literature; both were designed for clinical intervention studies. We instead used a lightweight, adapted checklist scored against four criteria, each rated 0–2 (absent/partial/clear): (Q1) clarity of the research question or system objective; (Q2) methodological or architectural transparency, i.e., whether the paper describes its design, pipeline, or algorithm in enough detail to assess or reproduce; (Q3) empirical validation, i.e., whether claims are backed by a reported evaluation (benchmark, user study, ablation) rather than narrative assertion alone; and (Q4) disclosure of limitations. Papers scoring 6–8 were rated *High* quality, 4–5 *Moderate*, 0–3 *Low*. Two reviewers scored each paper independently; disagreements greater than one point on any criterion were discussed to consensus.

No paper was excluded on quality grounds alone, since the survey aims to characterize the state of the field, including its immature and promotional corners, rather

Table 4 Quality-Assessment Summary by Source Type ($n = 40$)

Source type	n	**High**	**Moderate**	**Low**
Journal papers	8	6	2	0
Conference papers	23	14	9	0
Preprints (arXiv / SSRN working papers)	6	2	4	0
Institutional reports	1	0	1	0
Prototype/tool papers	2	0	0	2
Total	**40**	**22**	**16**	**2**

than only its strongest evidence. Instead, the quality rating is carried into the narrative: both papers rated *Low*, which are also the corpus's only two prototype/tool papers reporting self-measured, non-independently-replicated performance figures ([33, 34]), are flagged explicitly wherever their claims are cited, and their reported numbers are presented as self-reported rather than as validated findings (see the critical assessments in Section 7). Appendix A lists the per-paper coding and quality-assessment outcome for all 40 included studies, so the mapping from individual paper to quality band is fully transparent rather than aggregated only at the source-type level shown here.

2.6 Generative AI Use Disclosure

While preparing this manuscript, the authors used Claude (Anthropic, Sonnet 5.5) for language editing, mainly grammar correction, sentence restructuring, and clearer phrasing. The PRISMA 2020 reporting guidelines were also consulted when structuring the systematic review methodology, including study selection, screening, and reporting.

Claude was not used to produce research data or results, to carry out the final analysis, or to draw conclusions. The authors reviewed and edited all AI-assisted text themselves and take responsibility for the accuracy and originality of the final manuscript, as required by the Taylor & Francis Generative Artificial Intelligence (AI) Policy.

3 Definitions

The development of research-oriented AI systems has introduced several overlapping but distinct terms, including intelligent agents, multi-agent systems (MAS), and agentic AI. These terms are related but not identical; each emphasizes a different level of autonomy, interaction, and task execution. The surveys reviewed here show a clear shift from classical agent theory toward more flexible, tool-using, collaborative agentic systems [4–6].

3.1 Intelligent Agents

An *intelligent agent* is a computational entity that exists within an environment and acts autonomously toward specific objectives. Classical definitions say it perceives its surroundings, reasons about the situation, and selects actions that move it toward its

goals, so it is more than a reactive system. Falco and Robiolo associate agents with reactivity, proactivity, and social ability. The Belief-Desire-Intention (BDI) framework adds an internal cognitive structure for reasoning and action selection [4]. Such an agent should respond to changes in its environment, start actions without constant human intervention, and interact with other entities when cooperation is required.

Autonomy is central to the definition. A fully autonomous agent chooses actions from its internal state and context alone. A partially autonomous one still relies on human guidance or predefined policies. Recent work on agentic systems adds two traits: the agent adapts as new information arrives, and it communicates or coordinates with other agents or humans [5, 6]. In a research workflow, this allows an agent to search for literature, summarize evidence, compare sources, and revise its output across several steps instead of in a single pass.

3.2 Multi-Agent Systems

A multi-agent system (MAS) is a distributed system in which several agents share an environment and communicate with one another to solve problems that would be difficult or impossible for a single agent to handle alone. Agents can cooperate toward a common goal, compete for the same resources, or mix both behaviors depending on the task. Falco and Robiolo argue that MAS suit complex, dynamic environments because they support interaction modeling and distributed problem solving in ways centralized systems cannot [4].

MAS are not only a matter of distribution; interaction matters just as much. With multiple agents in the same system, the overall behavior can emerge from how agents interact, not just from what each one does on its own. That is why MAS work well for simulation, robotics, logistics, distributed decision-making [4], and healthcare. The same properties that make MAS powerful also make them hard to design and evaluate. Communication overhead, coordination failures, inconsistent goals, and unpredictability under uncertainty are the usual culprits.

3.3 Agentic AI

The term *agentic AI* describes a newer generation of AI systems that build on classical agent and MAS ideas by adding reasoning, planning, memory, external tool use, and collaboration to autonomy. Deshmukh et al. describes agentic AI as purpose-driven agents that can break tasks into simple steps, choose actions, and adjust its strategy over time [5]. Patil et al. find that recent agentic systems increasingly rely on modular architectures, communication layers, and coordination mechanisms, often built on large language models (LLMs) [6].

Traditional AI and agentic AI differ in how they operate. Conventional systems mostly handle prediction, classification, recommendation, or generation within a fixed input-output setting. Agentic systems perform sequences of actions, interact with external environments, and revise their strategy as new information arrives. For research support, that rarely means returning one response. The system runs a loop of retrieval, cross-source evidence comparison, and judgment about how well the available evidence supports an answer.

Recent surveys also describe agentic AI as increasingly hybrid. Symbolic components supply reliability and interpretability, and neural components supply flexibility and natural-language interaction [5, 6]. Research tasks need both.

3.4 Research AI Agents

Research AI agents are a subclass of agentic AI built for academic and research work. They support or automate tasks that researchers have traditionally done by hand: literature retrieval, evidence synthesis, citation analysis, summarization, comparative evaluation, and knowledge discovery. Unlike conventional search engines or static recommendation systems, they keep context across interactions and adjust their behavior as research needs and goals change [7, 8].

Recent literature increasingly treats research-oriented AI systems as autonomous processes instead of isolated models. Sami et al. point out that systematic literature reviews are slow because they require repeated searching, filtering, screening, and synthesis of large volumes of academic material [8]. Their proposed multi-agent architecture gives each stage of the review to a specialized agent: query generation, inclusion and exclusion filtering, abstraction and summarization, and final analytical synthesis.

Because they include reasoning and memory, modern research AI agents handle harder tasks than earlier retrieval-only systems. According to Akuma et al., contemporary agents combine planning, memory, and problem-solving with multi-step reasoning, external tool integration, and task management [9]. This lets a research agent find information across long, unstructured texts and revise its research questions mid-process. It can also compare sources over a long interaction.

Distinguishing factor between research AI agents and simple retrieval systems is how they access external resources without losing their reasoning thread. An agent can query an academic database, identify the main findings in matching papers, and assemble a working citation list, all within a single continuous reasoning loop without a sequence of prompts the user must re-enter each time [5, 7]. The underlying tool interfaces are not themselves new; databases, citation engines, and document-management repositories have been programmatically accessible for years. What is new is that the agent decides when to call them, evaluates what comes back, and folds that result into its ongoing reasoning.

Some architectures extend this with a self-reflection loop, in which the agent compares its own output against a quality standard before finalizing a response, and revises when the check fails [5]. Whether this consistently produces better results than a single well-prompted pass remains an open empirical question, but the architecture at least makes that kind of self-correction possible.

4 Use of AI in Research

AI now shows up throughout research workflows, where information volume is high, discovery moves fast, and manual processing is not feasible. Recent work describes a shift away from isolated task support and toward integrated systems: agents that assist with literature discovery and evidence synthesis, experimental planning, data analysis, scientific writing, and even peer review. [10–12].

Table 5 keeps two dimensions analytically separate. *Lifecycle stages* are points in the sequential, roughly chronological path a research project follows, from topic selection through dissemination; every stage row therefore corresponds to a segment of *when*, in a project's life, the AI capability is used. *Supporting activities* and *research methods*, by contrast, are cross-cutting: they can occur at several points in the lifecycle (funding applications recur; qualitative-analysis tooling can be invoked during literature review, writing, or analysis alike) and so are not tied to a single stage. Collapsing these two dimensions into one list, as earlier drafts of this table did, obscures which AI capabilities are stage-bound and which are used throughout a project.

Table 5: Major Uses of AI Across the Research Lifecycle, With Stages and Cross-Cutting Support Separated

Category	**Research Stage / Activity**	**Representative Systems / Studies**	**Primary Capability**	**Main Benefit**	**Key Limitation**
Lifecycle stage (sequential)	Stage 1: Topic exploration & ideation	ARC [1]	Topic identification, organization of early-stage work, and contextual recommendations	Lowers the barrier for novice researchers and supports early planning	Limited depth without human disciplinary guidance
	Stage 2: Literature review & knowledge extraction	Multi-agent SLR systems [8]; Semantic Scholar, Litmaps, Connected Papers, Research Rabbit [11]; materials synthesis framework [2]	Search-string generation, screening, extraction, semantic retrieval, and synthesis	Cuts manual effort in large literature reviews and supports structured evidence extraction	Depends heavily on retrieval quality; may miss nuance in complex sources
	Stage 3: Experimentation & analysis	AI4Science and agentic discovery systems [12, 16]	Hypothesis generation, experiment design, and workflow orchestration	Speeds up experimental planning and iteration	Reproducibility and verification remain significant challenges
	Stage 4: Scientific writing	ChatGPT-4 thesis assistant [13]; AI Scientist [10]; AI4Research [14]	Drafting, structuring, summarizing, formatting, and academic writing support	Speeds up drafting and gives writers more structure and confidence	Hallucination risk, overreliance, and need for human oversight

Category	Research Stage / Activity	Representative Systems / Studies	Primary Capability	Main Benefit	Key Limitation
	Stage 5: Peer review & evaluation	AI4Research review support [14]; AI Scientist review loop [10]	Reviewer-paper matching, pre-review/meta-review drafting, and quality-control checks	Speeds up screening and highlights obvious issues	Bias and verification remain significant challenges; not a substitute for expert judgment
	Stage 6: Dissemination & institutional governance	Institutional AI adoption frameworks [18]	Adoption planning, cross-disciplinary coordination, and policy alignment for research outputs	Gives institutions a structured path to responsible AI adoption	Governance and infrastructure maturity vary widely by institution
Supporting activity (cross-cutting)	Funding and proposal support	Transdisciplinary AI tools [15]	Call-text interpretation, team matchmaking, and proposal planning	Saves time and improves alignment with funding goals	Strategic and institutional judgment still depend on humans
	Qualitative-research tooling and document interaction	GPT-based qualitative workflows [17]; NotebookLM / ChatPDF-style interfaces [16]	Question answering over papers, conceptualization, and thematic analysis	Makes academic reading and synthesis more interactive	Fluent output can still be shallow, biased, or weakly grounded

4.1 Research Guidance and Topic Exploration

AI can also act as a personal research assistant. ARC, for example, helps undergraduate engineering students identify appropriate research topics, organize their work, and receive recommendations tailored to their educational environment [1]. Tools like this matter most for students who have completed relevant coursework but have not yet worked with an expert or research partner: their gap is often about not knowing where to take what they already know, not about a lack of ability.

The benefit is that AI can connect general-purpose academic tools to the needs of an individual student. Rather than offering generic suggestions, the system reads a student's profile and interests and makes recommendations calibrated to their current level of understanding [1].

4.2 Literature Search, Review Automation, and Knowledge Extraction

Literature review automation is one of the older applications of AI in research. Sami et al. show that a multi-agent system can automate most of a literature-review pipeline: generating search strings, applying inclusion and exclusion criteria, summarizing abstracts, and producing a final analysis for predetermined research questions [8].

Paper discovery and knowledge mapping are a related use. Kamath et al. describe tools such as Semantic Scholar that help researchers find relevant papers, visualize citation networks across studies, and surface thematic links between them [11]. Tracing citations by hand through databases takes time, and these tools cut it down. They can also reveal relationships that a simple keyword search would probably miss.

AI also helps with data analysis and predictive modeling. The same authors list environments such as Google Colab, Tableau, Power BI, RapidMiner, Weka, H2O.ai, and Scikit-learn, which make it easier to analyze data sets, build predictive models, and produce visualizations [11]. Here AI is a computational aid, not an independent researcher. It speeds up the analysis, and the researcher still does the interpretation.

Some systems let users converse with academic documents. Kamath et al. report a generative AI application with an integrated chat feature, built on tools such as HuggingFace and LangChain, in which users talk directly with a PDF and ask questions about a paper instead of reading it end to end [11].

Literature mining has more specialized systems. Lian et al. present an automated framework for intelligent synthesis design. It filters papers by semantic similarity, then uses progressive prompting and a dual-agent reasoning approach to extract key experimental details from materials-science literature [2]. The output is structured data from papers that were never written to be machine-readable.

4.3 Scientific Writing, Peer Review, and Experimentation

Some recent work covers more ambitious workflows. Tie et al. describe the "AI Scientist," a system designed to simulate the scientific process end to end, from idea generation through experimentation, writing, and paper generation [10]. Their survey finds working systems for each of these stages on its own. A fully automated pipeline that covers all of them has not been achieved.

The scientific-discovery literature is just as broad. Gridach et al. report AI agents used throughout the research process: literature review, hypothesis formulation, experimental design and execution, scientific writing, and paper generation. The fields covered include chemistry, biology, materials science, and engineering [12].

AI is also used for manuscript drafting and peer review. The AI4Research survey treats scientific comprehension, academic surveying, scientific discovery, academic writing, and peer review as interrelated processes, not isolated tasks [14]. On the writing side, AI can help draft and edit manuscript sections, improve readability, summarize results, and format submissions to journal requirements. On the review side, it can match papers to reviewers and handle pre-review or meta-review tasks. Whether any of this shortens time to publication or only moves the bottleneck is not yet clear.

Systems can extract knowledge from scientific texts and multimodal artifacts, identify concepts, structure document content, and give concise answers to specific questions about scientific material [14]. This matters because much of what is useful in a paper, a specific number in a table or a trend buried in a figure, is exactly what plain-text search tends to miss.

4.4 AI for Research Funding and Proposal Support

Research funding preparation is now a major workload in academia: writing proposals means interpreting call texts, aligning with strategic policy goals, and responding to requirements on ethics, sustainability, open science, interdisciplinarity, and impact. Spyroglou et al. argue that this creates substantial pressure on researchers and administrators, especially when success rates are low and funding instruments are complex. They propose transdisciplinary AI tools to support matchmaking, team composition, conceptualization, and concretization of research plans [15].

AI can help researchers navigate the large amount of information tied to modern calls for proposals by summarizing requirements, identifying thematic alignment between a proposal idea and a funding call, and cutting the time spent on document inspection. The interpretation of funding priorities, negotiation of consortium roles, and assessment of strategic fit still depend on researcher judgment and institutional knowledge [15].

4.5 Challenges and Limitations

The same limitations keep appearing across the literature. AI-generated content can hallucinate, lack authenticity, rest on weak factual foundations, and be hard to replicate, especially in systems that run several steps without human checks [7, 12]. Ethical concerns are just as common: transparency, citation integrity, bias, and responsible verification. Citation integrity matters most for research. A hallucinated citation in a paper does more damage than a hallucinated fact in a chatbot reply.

Chubb et al. find that researchers see AI as useful for gathering information and handling small tasks. They also see it as part of a wider trend toward heavier workloads and pressure to meet academic productivity metrics [19]. The same study flags risks of bias in peer review, loss of nuance, and AI-driven research infrastructure turning into a tool for surveillance or algorithmic management of researchers themselves.

Qualitative-research studies raise a related caution. AI-generated content can sound fluent but shallow, and it may contain fabricated or misleading information [17]. Christou recommends that researchers get familiar with AI output, check it for bias, and verify it against other data sources. The analysis itself should stay under human control, with clear evidence of the researcher's own cognitive input throughout [17]. These guidelines carry over to research AI agents, where a human has to stay involved from the first step to the last.

4.6 Semantic Discovery and Document-Level Question Answering

Recent surveys report that AI-supported literature discovery now goes well beyond keyword matching. Researchers get something closer to a map than a ranked list. Semantic retrieval and citation-aware exploration let them navigate by meaning and by the relationships between papers, so a paper can turn up even when it lacks the search terms [16].

NotebookLM and ChatPDF-style paper-chat systems let researchers load a small set of relevant papers and ask for targeted answers or summaries [16]. Retrieval-augmented generation improves usability and lowers hallucination risk compared with purely generative systems. Quality still depends on the retrieval strategy and the grounding.

4.7 Scientific Experimentation and Idea Generation

AI is used for scientific idea generation and experimentation. Eger et al. group this under AI4Science, which covers AI-assisted hypothesis generation, experiment design, and automated experimentation [16]. It differs from literature processing. That work summarizes what has been tried and this proposes what to test next.

Hypothesis generation systems look for patterns across literature, data, and prior experiments, then suggest candidate hypotheses [16]. They link patterns across documents and datasets that would take much longer to assemble by hand. They also help with early planning by cutting a large design space down to a few directions worth testing.

Experimentation support is more direct. AI can help with code generation, parameter selection, and workflow orchestration, which reduces repetitive work and shortens iteration cycles [16].

4.8 AI for Qualitative Research Practice

Christou notes that AI has been used in qualitative inquiry, including literature and systematic reviews, conceptualization, and thematic and content analysis. These uses raise concerns about unreliability, bias, and unethical behavior [17]. The concerns are sharper than in quantitative work, because qualitative research depends on interpretation and context, and AI systems tend to flatten both.

Christou argues that GPT-based systems can help produce summaries, generate research questions, and support early-stage conceptualization, but that their output should never be taken at face value [17]. Qualitative research is about interpreting meaning, not sorting content into categories. Fluent but shallow output is therefore a particular risk, since it can pass for analysis without doing any interpretive work.

4.9 Institutional and Policy Perspectives on AI in Research

The Guild's position paper on AI in research sets out the opportunities and risks of AI in universities and research institutions, along with the governance needed to adopt it responsibly [18].

The report argues that AI is now widespread in academia and affects researchers, students, educators, and administrative staff [18]. It names literature review, hypothesis formulation, data analysis, visualization, experimentation, and dissemination as areas where AI can help. Alongside this, it calls for openness, transparency, multilingual support, and cross-disciplinary collaboration.

Taken together, the literature suggests AI now plays several roles in research: it supports research initiation through personalized guidance, enables large-scale literature mining and structured synthesis, automates or semi-automates systematic literature review workflows, and shapes the broader culture of research through productivity expectations and institutional practices [1–3, 19]. Systems described as hybrid, combining automation with a human-in-the-loop checkpoint, are more often reported as working well than fully autonomous designs (see Table 6 for the specific comparisons this rests on). Across sources, the recurring theme is that AI works best when it augments human scholarship rather than replacing it.

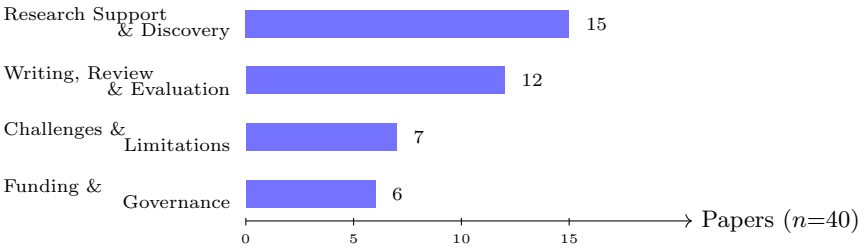


Fig. 2 Thematic distribution of the 40 reviewed papers by primary focus area. Counts reflect the dominant coding category per paper; papers may address secondary themes not shown.

As shown in Fig. 2, the reviewed corpus is dominated by research support and discovery themes (15 papers), followed by scientific writing, peer review, and evaluation (12 papers), while funding/governance (6) and challenges/limitations (7) receive comparatively less coverage.

5 Architectures

The architecture of research AI agents sits at the intersection of classical agent theory, large language models, retrieval pipelines, and multi-agent orchestration. The current literature no longer converges on a single design pattern. Instead, it points to a family of architectural choices that differ in autonomy, coordination, grounding, and deployment style. The Concept-Code-Commerce lens is a useful starting point: it places AI agents across theoretical foundations, implementation technologies, and societal deployment, and in doing so shows that architecture must balance reasoning capability with practical usability [20].

An earlier version of this survey listed these architectural items, retrieval-augmented generation, multi-agent coordination, BERT-based evaluation models, AgentOps, and institutional integration, side by side, as if they were mutually exclusive choices at the same level of abstraction. They are not: they answer different questions,

and a single deployed system typically combines one item from each level. We therefore organize the remainder of this section, and Table 10, around four taxonomic levels:

- **Level 1 – Component techniques.** Building blocks used *inside* an agent to perform a specific function: retrieval-augmented generation (Section 5.2) for grounding, and task-specific fine-tuned models such as BERT-based evaluators (Section 5.1) for scoring or classification. A component technique is not itself a complete agent.
- **Level 2 – Agent architecture patterns.** How an individual agent, or a coordinated set of agents, is structured and makes decisions: classical agent models, LLM-based agents, Agentic RAG, Multi-Agent Autonomous Workflow Systems (MAWS), TacTool-style tool-calling separation, and closed-loop autonomous-discovery architectures (Sections 5.1–5.5). This is the level most prior surveys treat as the only level.
- **Level 3 – Operational management layer.** Cross-cutting infrastructure for running and maintaining an agent *regardless of which Level-2 pattern it uses*: AgentOps-style observability, monitoring, and automated tuning (Section 5.5).
- **Level 4 – Deployment and institutional integration.** How an agent architecture is adopted at organizational scale: enterprise governance layers and library/institutional integration frameworks (Section 5.7). This level is about organizational readiness and governance, not about how the agent itself reasons.

Table 10 makes this level assignment explicit for all twelve architecture-related items identified in the corpus, which keeps the comparison honest: it no longer implies, for example, that RAG (a Level-1 technique) and enterprise governance (a Level-4 deployment concern) compete for the same design decision.

5.1 Classical Foundations and LLM-Based Agent Architectures

Early architectural ideas in AI agents centered on autonomy, decision-making, and coordination. Ishida et al. trace these ideas to foundational work on the Blackboard Model and the Contract Net Protocol, which established early mechanisms for distributed problem solving and negotiated task allocation among agents [20]. The main classical families within autonomous agent research are reactive, deliberative, layered, and learning-based models.

Large language models have reshaped agent architecture by adding natural language reasoning, instruction following, and tool orchestration to the traditional autonomy stack. Chen and Peng describe an LLM-based AI agent as a system built around four core components: the LLM itself, a memory module, task-planning capability, and a tool-usage mechanism [21]. Separating language understanding from state retention, planning, and external action makes this architecture useful.

Prompt engineering is a central design element in these architectures. Chen and Peng argue that prompts form the control layer connecting user intent to the agent's reasoning process [21]. In practice, the most workable LLM-based architecture is a modular system where memory, planning, tools, and reasoning are handled separately [20, 21].

5.2 Retrieval-Augmented Architectures (Level 1: Component Technique)

Retrieval-augmented generation is a common architecture for research-oriented agents as it pairs language generation with grounding in external sources. Han et al. describe RAG as a three-stage pipeline: retrieval, augmentation, and generation [22]. This reduces reliance on static pretraining and limits hallucination when a research problem needs up-to-date or source-backed answers.

This architecture matters for systematic literature review automation. Han et al. propose an RAG-based framework covering four core SLR stages: literature search, literature screening, data extraction, and information synthesis [22]. RAG makes the research agent more verifiable as the agent can point to retrieved scientific sources.

Sparse retrievers such as BM25 are efficient and interpretable, but they lean heavily on keyword overlap. Dense retrievers capture semantic similarity and handle nuanced academic queries better, though they cost more to run. Search engines can also work as retrievers when the system needs current information [22].

5.3 Multi-Agent Autonomous Workflow Architectures (Level 2: Agent Pattern)

A second architectural pattern uses multiple autonomous agents that work together through explicit workflows. Apuri and Yepuri define Multi-Agent Autonomous Workflow Systems (MAWS) as systems in which autonomous agents collaborate, reassign tasks dynamically, and adapt to changing conditions in their environment [23]. Centralized workflow management is rigid. MAWS distributes authority instead, and the agents coordinate around shared objectives without a fixed central schedule.

A distinguishing feature of MAWS is agent specialization and the communication links between agents. Each agent can make independent decisions based on the information it receives, inform other agents, and adjust its behavior in response to environmental conditions [23]. The paper also reports using methods such as deep reinforcement learning and collaborative multi-agent learning to improve policy choice and coordination, alongside collaborative planning and hierarchical task organization as further architectural patterns.

5.4 Agentic RAG: From Traditional RAG to Adaptive Pipelines (Level 2: Agent Pattern)

Agentic RAG is an answer to the limits of standard retrieval-augmented generation [27]. Traditional RAG systems follow a largely linear, static path. Agentic RAG adds decision-making around the retrieve-augment-generate cycle, so the pipeline can adjust as it runs.

Singh et al. identify four core agent components that make this possible: the LLM, memory, planning, and tools. Together these make the agent stateful and adaptive rather than stateless and reactive [27]. They also identify four recurring agentic workflow patterns, reflection, planning, tool use, and multi-agent collaboration,

implemented through common workflow structures such as prompt chaining, routing, parallelization, orchestrator-worker systems, and evaluator-optimizer designs.

The taxonomy starts with the single-agent architecture, where one coordinating agent handles query analysis, source selection, retrieval, and response synthesis. The multi-agent extension separates coordination from retrieval and synthesis: a master retrieval agent delegates to specialized agents [27]. Graph RAG adds graph structures to represent relationships among entities, concepts, and documents, which enables multi-hop reasoning and contextual enrichment.

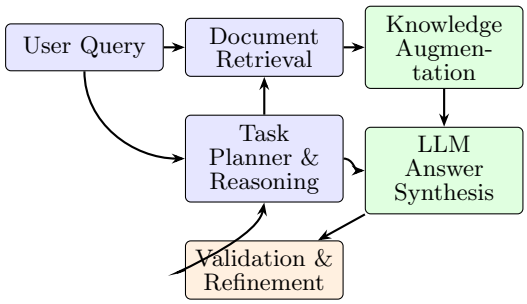


Fig. 3 Agentic RAG pipeline for systematic literature review. Adapted from [27]. The system extends traditional RAG with iterative feedback loops, multi-step reasoning, and adaptive retrieval, enabling context-aware knowledge synthesis and validation.

5.5 Tactical Tool-Calling (Level 2) and AgentOps (Level 3)

Reliable tool usage is a central challenge in agentic AI architecture. Singh et al. note that even strong models can fail at reasoning about whether a tool is needed or they fail at formatting the call itself [28]. TacTool separates these two problems: one LLM reasons and produces a concise plan, while a second LLM calls the tool and executes the function. A response-selection layer then compares the two outputs using an LLM-as-a-judge mechanism, so the system can recover from partial failures [28]. TacTool is a Level-2 agent pattern: it changes how a single agent decides to act.

AgentOps, by contrast, sits at Level 3: it does not change how an agent reasons. It wraps around whichever Level-2 pattern is deployed to keep it observable and maintainable at runtime. Autonomy brings operational uncertainty. Moshkovich and Zeltyn argue that agentic systems are inherently probabilistic: they depend on LLM reasoning, changing memory states, shifting tool availability, and execution paths that evolve at runtime [29]. Their AgentOps framework runs as a six-stage pipeline: observe behavior, collect metrics, detect issues, identify root causes, generate recommendations, and automate operations.

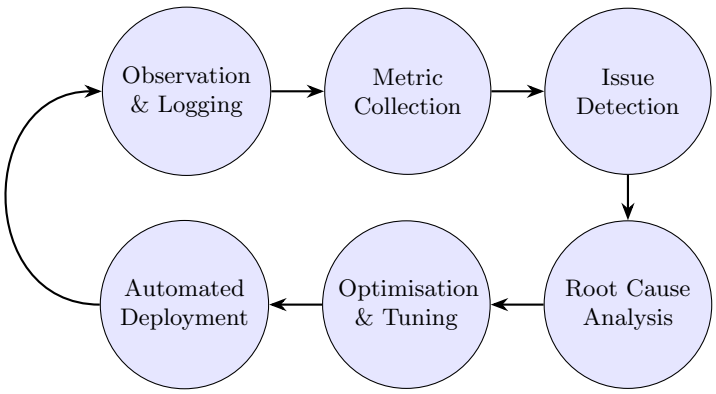


Fig. 4 AgentOps lifecycle for managing uncertainty in agentic AI systems. Adapted from [29]. The pipeline forms a continuous feedback loop for system reliability and adaptation.

5.6 Autonomous Scientific Discovery and Layered NLP Architectures (Level 2), and BERT-Based Evaluation (Level 1)

Kohli et al. describe autonomous AI agents as systems capable of planning, learning, experimentation, reasoning, and adaptation inside closed-loop scientific workflows [24]. They identify five coupled capabilities underlying this architecture: perception, planning and reasoning, learning, execution, and adaptation. One concept that follows from this work is the "self-driving laboratory," where AI planning systems connect directly to robotic experimental infrastructure. Closed-loop systems of this kind raise open questions about reproducibility, interpretability, trust, and safety [24]. Like MAWS and Agentic RAG, this is a Level-2 agent pattern.

Sharma et al. report that, on their text-classification benchmark, an agentic architecture combining adaptive learning, memory retrieval, multi-agent collaboration, and reinforcement-based refinement outperformed the static classifiers they compared it against [25]; this is a study-specific, single-paper comparison rather than a corpus-wide finding. Their architecture has five layers: Perception, Memory and Knowledge Base, Decision Engine, Actuation, and Feedback and Learning, again a Level-2 pattern. Chen takes a narrower approach and proposes a BERT-centered model for academic writing evaluation. The model splits assessment into preprocessing, contextual embedding, specialized classifiers, holistic scoring, and feedback generation [26]. This is a Level-1 component technique. It is a fine-tuned scoring model that could sit inside a larger agent, but it is not an agent architecture itself. We therefore treat it separately from the Level-2 pattern of Sharma et al., although both papers use the word "architecture."

5.7 Enterprise-Scale and Library Integration Architectures (Level 4: Deployment)

According to Singh et al., agentic workflows add autonomous cycles of refinement, planning, and collaboration before the final response is produced [30]. The common

patterns are reflection, tool use, planning, and multi-agent collaboration. Pandey and Narnag build on this with the Integrated Agentic Foundry (IAF), a governance-first architecture with four layers: foundational (multimodal knowledge graph), cognitive orchestration, embodied perception-and-action, and governance-and-interoperability [31]. The governance layer adds standardized protocols such as Model Context Protocol (MCP) and Agent-to-Agent (A2A) communication.

Ivanova's engineering framework covers the integration of AI into research and academic library infrastructures [32]. It divides adoption into three stages: organizational preparation, implementation of AI-ready solutions, and integration of external AI expertise. The paper cites UNESCO's ethical standards and the European Union AI Act as guidance.

5.8 Architectural Trade-Offs and Comparative Analysis

Full comparison in Supplementary Table 10.

Table 10 summarizes the twelve architecture-related items reviewed, now organized by the four taxonomic levels introduced above rather than listed as a single flat inventory. No single Level-2 pattern is universally optimal. Lightweight LLM agents are flexible but may lack grounding. RAG systems improve factual reliability but depend heavily on retrieval quality. Multi-agent systems increase scalability but introduce coordination overhead. Enterprise-scale and institutional architectures provide governance and interoperability but require substantial infrastructure [31, 32].

For research AI agents specifically, hybrid modular architectures, those combining LLM reasoning, retrieval grounding, memory systems, tool orchestration, and observability mechanisms into one coordinated framework, are more frequently reported as outperforming monolithic or single-technique designs. Table 6 collects the specific within-paper comparisons this claim rests on. We are explicit about what this evidence does and does not show: each row is a comparison reported *within one primary study*, typically against a baseline the same authors defined, not a controlled, head-to-head benchmark run across architectures by an independent third party. No paper in the corpus benchmarks two or more of these architecture families against each other under identical tasks and metrics. The pattern is therefore best read as a convergent signal across independent research teams rather than as settled comparative proof, which is why we phrase it as a frequently-reported finding rather than a validated result.

6 The Proposed Architectural-Lifecycle Framework

The abstract of this survey claims an original architectural and lifecycle-based framework. We make that framework explicit here rather than leaving it implicit in the preceding two sections, and we state directly how it differs from the three closest prior surveys.

6.1 Structure of the Framework

The framework has two axes, developed independently in Section 4 (the six-stage lifecycle, Table 5) and Section 5 (the four-level architectural taxonomy, Table 10). Its

Table 6 Within-Study Comparative Evidence Cited for the Hybrid-Modular Pattern

Source	Comparison reported	Nature of evidence
[25]	Layered agentic architecture (perception, memory, decision, actuation, feedback) outperformed static classifiers on the authors' own text-classification benchmark	Single-paper benchmark, author-defined baseline
[28]	Two-LLM TacTool design (separate reasoning and execution roles) reduced malformed tool calls relative to a single-LLM tool-calling baseline	Single-paper ablation, author-reported
[22]	RAG pipelines reduced unsupported/hallucinated claims relative to generation without retrieval grounding, across the SLR studies surveyed	Narrative synthesis across multiple primary studies, not a new head-to-head experiment
[33]	Multi-agent CrewAI pipeline reported higher accuracy and lower processing time than the authors' own non-agentic baseline pipeline	Single-paper, self-reported; rated *Low* quality (Section 2.5) – included here for completeness, not as independent corroboration

Table 7 Architectural Statistics Extracted from the Reviewed Literature

Architectural Element	Count	Description
Classical agent families	4	Reactive, deliberative, layered, and learning-based architectures.
Core LLM-agent components	4	LLM, memory, planning, and tool-usage modules.
RAG pipeline stages	3	Retrieval, augmentation, and generation.
SLR workflow stages	4	Literature search, screening, data extraction, and synthesis.
Agentic workflow patterns	4	Reflection, planning, tool use, and multi-agent collaboration.
AgentOps operational stages	6	Observation, metric collection, issue detection, root-cause analysis, optimization, and automation.
Layered text-classification modules	5	Perception, memory, decision engine, actuation, and feedback-learning layers.

contribution is to cross them: for each lifecycle stage, it identifies which architectural level currently carries the AI capability used at that stage, and which levels are absent. This answers a question neither axis answers alone: not just "what does AI do at this stage" (the lifecycle table) and not just "how is an agent built" (the architecture table), but "which class of architecture is actually responsible for supporting this stage today, and is that support structurally adequate or merely incidental."

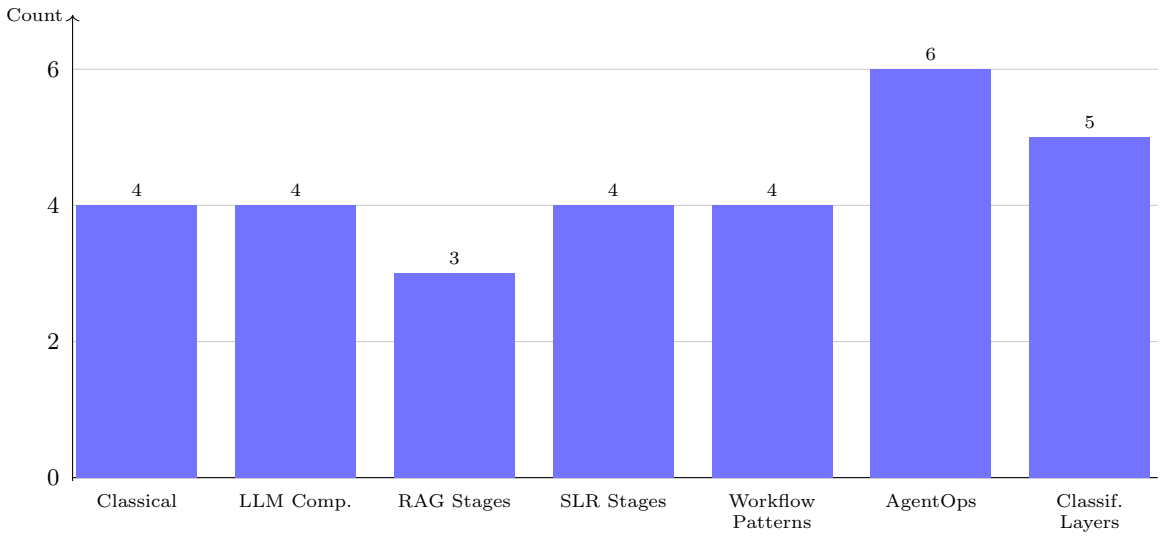


Fig. 5 Architectural patterns and workflow structures identified in the reviewed AI-agent literature. Surveyed systems consistently emphasise modular architectures, retrieval grounding, planning, multi-agent coordination, and operational observability for literature-review automation.

Each stage×level cell in Table 8 is rated against one criterion: *how many of the reviewed papers report a system at that level explicitly targeting that lifecycle stage, and how developed is that reported system.* We apply this consistently using four bands: *Absent* means no paper in the corpus reports a system at that level for that stage; *Weak* means one paper reports a system, but it is a narrow or repurposed use (e.g., a general-purpose LLM agent applied to the stage without a stage-specific design) rather than a purpose-built pattern; *Partial* means two or more papers report systems, but they are early-stage, evaluated only informally, or address only part of the stage's tasks; *Strong* means three or more papers report systems that are purpose-built for the stage and include at least some reported evaluation (benchmark, case study, or user study, per the Q3 criterion in Section 2.5). The same two reviewers who conducted the coding in Section 2 assigned these bands independently, using the coding sheet's lifecycle-stage and architectural-pattern fields, and resolved disagreements by discussion.

Table 8: The Architectural-Lifecycle Framework: Cross-Mapping of Lifecycle Stage Against Architectural Level

Lifecycle Stage	**L1: Technique present**	**L2: Agent pattern present**	**L3: Operational layer present**	**L4: Deployment layer present**	**Structural gap**
1. Topic exploration & ideation	Weak (profile-based recommendation only)	No dedicated pattern; single-turn LLM use	Absent	Absent	No agent architecture yet targets this stage specifically
2. Literature review & knowledge extraction	Strong (RAG is the default grounding technique)	Strong (Agentic RAG, MAWS, multi-agent SLR systems)	Partial (AgentOps applies but is rarely reported *for* this stage)	Weak	Best-covered stage overall; operational monitoring is the main gap
3. Experimentation & analysis	Weak (task-specific models, not yet standardized)	Partial (autonomous-discovery patterns exist but are early-stage)	Absent	Absent	Closed-loop safety and reproducibility controls are largely undeveloped
4. Scientific writing	Partial (BERT-based evaluators for scoring)	Partial (LLM-based agent pattern, not writing-specific)	Absent	Absent	No writing-specific agent pattern; relies on general LLM agents
5. Peer review & evaluation	Weak	Weak (folded into AI4Research as a task, not a dedicated pattern)	Absent	Absent	Least architecturally developed stage in the corpus
6. Dissemination & institutional governance	Absent	Absent	Absent	Strong (Enterprise Agentic, Library AI Integration)	Governance frameworks exist but are decoupled from the agent patterns used earlier in the pipeline

6.2 What the Framework Shows

Two patterns stand out. First, architectural maturity is concentrated at Stage 2 (literature review) and, separately, at Level 4 (institutional deployment); these two areas

of strength do not overlap with each other, since the enterprise and library-integration architectures reviewed in Section 5.7 are described at the organizational level and are not reported as wrapping the literature-review agent patterns reviewed in Section 5.4. Second, Stages 1, 3, and 5 (ideation, experimentation, and peer review) each lack a dedicated Level-2 agent pattern: the AI capability used there is either a general-purpose LLM agent repurposed for the task, or, in the case of experimentation, an early-stage closed-loop system whose safety and reproducibility properties are still open questions [24]. No paper in the corpus proposes an AgentOps-style operational layer (Level 3) for any stage other than literature review, which means runtime monitoring, once a system is deployed for writing support, experimentation, or peer review, is largely unaddressed by the reviewed literature.

6.3 Differentiation From Prior Surveys

This cross-mapping is what distinguishes the present framework from the three closest prior surveys, extending the comparison introduced in Section 1. Bolaños et al. [3] analyze architectural choices only within Stage 2 and do not extend the comparison to other lifecycle stages or to Levels 3–4. Gridach et al. [12] survey agentic techniques across several stages but do not organize those techniques into the leveled taxonomy used here, so their survey cannot distinguish, for example, a Level-1 technique from a Level-4 deployment concern. Chen et al. [14] catalog AI4Research tasks stage-by-stage but treat architecture as incidental detail rather than as a second, independently structured axis. None of the three produces a table analogous to Table 8 that lets a reader see, at a glance, which lifecycle stages are architecturally underserved.

7 Existing Tools

7.1 General-Purpose LLM Assistants

7.1.1 ChatGPT-4 for Thesis Development

In the case study reviewed, ChatGPT-4 was tested as an academic assistant for postgraduate thesis development, supporting definition of the research problem, the literature review, and formulation of the research hypothesis [13].

NotebookLM is a retrieval-based note-taking tool, developed by Google Labs (built on Google's Gemini models) rather than by Google DeepMind, where users upload their own source documents instead of searching the open web and query directly against their own notebooks. Hallucination risk is reduced by this closed-corpus Q&A design, since answers are bound to the uploaded content rather than to a general chat interface. Eger et al. categorize NotebookLM under a broader umbrella of paper-chat tools designed for focused academic reading [16]. The downside is that the tool's scope is limited to what the user uploads, so it is not well-suited to a thorough literature review on its own.

Elicit is an AI research assistant that automates part of the literature review. It searches for papers, pulls out their key claims, and fills in structured comparison tables.

Verified against Google's own product documentation at https://notebooklm.google as of September 2026; an earlier draft of this survey misattributed development to Google DeepMind.

The user poses a research question, and Elicit returns a ranked list of papers with the methodology, findings, and limitations of each. The output fields and extraction templates are built to align with systematic-review conventions. Elicit's own documentation describes its corpus as built primarily on the Semantic Scholar dataset rather than as a single-source index; even so, coverage is uneven across disciplines (weighted toward biomedical and social-science literature), and extraction quality drops for specialized or interdisciplinary topics that the underlying corpus covers less well. Coverage claims for commercial tools such as Elicit change without notice and should be re-verified against the vendor's current documentation before publication.

7.1.2 Consensus

Consensus is a search engine for scientific papers that can produce a consensus based on evidence rather than keyword matching at the study level. It pulls apart relevant papers and generates a consensus meter that rates the support or rejection of a specific claim in the literature, useful for evidence-checking or to support a hypothesis before further digging. There's one big problem with the meter, however: it collapses all the research, irrespective of design and sample size, and fails to account for the quality of the research; as a result, it can give the impression of a more stable or definitive summary than the actual study results.

7.2 Literature Discovery Platforms

7.2.1 Semantic Scholar

Semantic Scholar, from the Allen Institute for AI, is a large-scale academic search engine built on NLP-based semantic indexing and citation analysis. It handles semantic search, paper recommendations, citation graph navigation, and TLDRs, one-sentence summaries of each paper. Kamath et al. describe it as useful for mapping knowledge and spotting thematic relationships between papers [11]. It also quietly does a lot of heavy lifting behind the scenes: several multi-agent tools, including OmniScholar and ScholarGenie, use it as their retrieval backbone. Semantic Scholar itself does not support interactive synthesis or structured extraction natively.

7.2.2 Connected Papers

Connected Papers builds a visual citation network around a seed paper, letting researchers navigate related work through graph visualization instead of keyword search. Each node is a paper, each edge a citation relationship, and node size tracks citation count. Kamath et al. note its value for surfacing thematically related papers that might not share a single keyword with the seed [11]. The tool earns its keep in the exploratory phase of a literature review, but that's where it stops: no extraction, no summarization, no synthesis.

7.2.3 Research Rabbit

Research Rabbit is a free citation-graph tool that turns a seed set of papers into an interactive map of related literature. What sets it apart from Connected Papers

is that the network isn't fixed. Users can keep refining it, adding papers, dropping ones that turn out to be dead ends, and the tool also plugs into Zotero for reference management. The downside shows up at scale: feed it a large seed set and the graph gets tangled fast, and there is no automated summarization to fall back on.

7.2.4 Litmaps

Litmaps generates chronological citation maps of how a research area has developed: when the key papers appeared, how later work cited them, and where gaps remain. Kamath et al. describe it as a tool for detecting thematic relationships across studies [11]. It tracks research trajectories and points out influential work, but it stops there. The researcher still does the extraction and synthesis.

7.2.5 Scite.ai

Scite.ai labels citations as supporting, contradicting, or merely mentioning a claim, which says more than a raw citation count. Researchers can use these "smart citation" labels to see whether a finding has been replicated or challenged, or only cited in passing. Systematic reviewers can use them to judge how much weight a claim can bear. Accuracy varies by domain, and coverage is thinner for older and non-English literature.

7.3 Multi-Agent Synthesis Systems

7.3.1 OmniScholar for Researcher Profiling and SDG Mapping

OmniScholar is a tool for automatically extracting, structuring, and interpreting the bibliographic metadata of a researcher's publications [33]. Google Scholar, ResearchGate, and Semantic Scholar are all large stores of publication data, but none of them automate the full process of profiling an author, filtering their work, and correlating it to thematic objectives such as the UN Sustainable Development Goals (SDGs); that is the gap OmniScholar is designed to fill.

The authors compare two architectures head to head: a multi-agent CrewAI pipeline, with separate agents for retrieval, Gemini-based formatting, and TF-IDF SDG mapping, versus a non-agentic pipeline that queries Semantic Scholar, CrossRef, and other sources directly through a custom API process [33]. The highest accuracy reported was 92% for keyword-based SDG classification, and a ranking system integrates citation counts, publication dates, author relevance, and SDG similarity into a single composite score. The paper also reports a time saving of more than 95% relative to manual profiling.

Critical Assessment.

As flagged in the quality assessment (Section 2.5, Table 4), this paper is rated *Low* quality: the numbers come from a single conference paper with no independent evaluation or adversarial testing. The 92% accuracy figure is measured against a dataset the authors themselves constructed, and the 95% time-reduction figure is not accompanied by any described control condition. The scraping component is also fragile:

a change to a source website's structure, or an ambiguous author name, can break it. The output is interesting but unbenchmarked and should be treated as indicative rather than validated; we recommend manual verification of OmniScholar's output before using it for institutional reporting.

7.3.2 ScholarGenie for Automated Literature Summarization and Presentation Generation

ScholarGenie is a modular multi-agent framework built on CrewAI and LangChain. A single pipeline covers literature retrieval, semantic summarization, and slide generation [34]. It consists of an Autonomous Literature Agent, a Semantic Contextual Summarizer, a GenieSlide Generator, and an optional text-to-speech agent. Before anything reaches the summarizer, retrieval goes through topic expansion, synonym rewriting, and relevance ranking. The summarizer then splits each paper into problem statement, methodology, key findings, and limitations. The paper reports that literature processing takes up to 80% less time and that slide decks are generated in under a minute.

Critical Assessment.

ScholarGenie has the same weakness as OmniScholar and receives the same *Low* quality rating (Table 4). Its results come from a single conference paper that no one has independently replicated. The 80% time-reduction figure is based on self-reported comparisons by users, and no controlled experiment backs it up. Even the authors note that output quality depends heavily on retrieval quality and prompt specificity, and that summarization can flatten highly specialized topics into something simpler than they really are. ScholarGenie is best read as a prototype that demonstrates the agentic-synthesis pattern, not as a production-ready research assistant.

Full comparison in Supplementary Table 11.

8 Ethics and Security

Ethics and security issues with AI in research have become a main concern. They are now the focus of the literature. There are two themes in the recent work. First, the principles of AI systems, transparency, explainability, accountability, privacy, equity, and responsible verification of AI-generated knowledge should be researched and regulated. Secondly, agentic systems present a new kind of security surface. The synergy of language models and memory, planning, and tool utilization provides space for prompt injection, jail breaks, backdoors, privacy leakage, and unsafe tool executions [35–38].

8.1 Ethical Principles for AI in Education and Research

Educational and research contexts have raised a small number of common concerns in the systematic reviews of AI ethics that have been conducted: transparency/explainability, privacy and data protection, responsibility and accountability,

and equity. Themes come up repeatedly for a reason. In the field of research and education, AI systems are not merely background technologies—they have a profound impact on the decisions made that impact learners, researchers and institutions [35].

8.2 Responsible Use of AI in Scientific Research

Rojas Contreras and Portilla Jaimes also critique a limited understanding of ethics as mere adherence to rules. Their broader view includes AI literacy, stakeholder engagement, responsible research practices, inclusion and diversity, and AI-generated knowledge verification. The paper draws on UNESCO guidance and calls for consideration of ethics throughout the entire AI lifecycle.[35].

Resnik and Hosseini describe how AI can assist with many science research tasks, including data analysis, hypothesis generation, modeling, literature search, writing, and peer review. They note that these capabilities raise ethical concerns that current scientific norms do not explicitly address [36]. Rather than a new ethics for science, they call for supplementary AI-specific guidance on top of existing norms. Researchers should be aware of AI bias and error and work to minimize both. AI use should be disclosed in language that non-experts can understand and follow. Affected communities should be engaged whenever AI becomes part of the research process.

Synthetic data and authorship get specific treatment. Synthetic data should be identified, the process used to create it described, and its use made explicit instead of hidden. AI systems may be acknowledged as contributors, but their names must not be listed as authors, inventors, or copyright holders, and the extent of AI's contribution should be disclosed elsewhere in the work [36].

8.3 Opportunities and Risks of AI-Based Research Applications

The Ukrainian universities study reports what academic staff actually think about AI in research. The main benefits respondents cited were automation of repetitive work, data extraction from many sources, and faster processing of large amounts of data. They were less sure that AI makes research better, yields unexpected insights, or builds new skills. They also raised concerns about unreliable results, weaker research integrity, unethical use of AI in reviews, a greater need for peer audit, and more plagiarism [39].

Khlaif et al. evaluate ChatGPT on research tasks and find that reviewers judged its output potentially publishable when the prompts supplied enough context. The results were uneven. The effect was smaller for research-framework development and data analysis, and the literature-review section was the most demanding [40].

8.4 Security Threats in AI Agents

AI agents have a wider attack surface than ordinary AI tools because they combine reasoning, memory, and tool execution in one system. Li et al. treat agent security as a layered problem and sort vulnerabilities by the component they affect: the LLM itself, reasoning and planning, memory, tools, or multi-agent communication [37]. In

their view, securing the model is not enough. An attack can enter through any of these components, so the whole workflow needs protection.

8.5 Protective Measures and Operational Safeguards

For each layer, Li et al. propose a matching defense: prompt filtering, secure context handling, controlled tool access, and monitoring of multi-agent communication. They present these as practical engineering measures, and the aim is to stop one compromised component from taking down the rest of the system [37].

Secure agent design should also isolate user sessions, sanitize inputs before any tool runs, and limit which commands an agent can issue. Actions generated by the model should be treated as untrusted until they have been validated [38].

Table 9: Comparison of Ethical and Security Issues in AI-Based Research Systems

Paper / System	**Primary Focus**	**Main Ethical Concerns**	**Main Security Concerns**	**Suggested Mitigations**
Rojas Contreras & Portilla Jaimes (2024) [35]	AI ethics in education and research	Transparency, explainability, accountability, equity, AI literacy	Risks from uncontrolled AI deployment and misuse	Ethical governance frameworks, transparency policies, stakeholder participation
Resnik & Hosseini (2025) [36]	Ethical guidance for AI in scientific research	Bias, authorship ambiguity, synthetic data misuse, inadequate disclosure	Unverified AI-generated scientific outputs	Mandatory AI disclosure, labeling of synthetic data, human accountability
Borodiyenko et al. (2025) [39]	AI perceptions in Ukrainian universities	Research integrity, plagiarism, unreliable findings, overdependence on AI	Weak reliability verification, uncontrolled AI-generated information	Institutional policies, staff training, peer-review strengthening
Khlaif et al. (2023) [40]	ChatGPT performance in scientific writing	Ownership ambiguity, academic integrity, hallucinated literature reviews	Potential misuse of generated content	Human oversight, detailed prompting, critical evaluation
Li et al. (2025) [37]	Security analysis of AI agents	Unsafe autonomous behavior, manipulation of agent reasoning	Prompt injection, jailbreaks, backdoors, memory poisoning, tool misuse	Prompt filtering, secure memory, controlled tool access, layered defense
He et al. (2025) [38]	Security architecture and operational risks	Trustworthiness of autonomous agents	Privacy leakage, session hijacking, model pollution, availability attacks	Session isolation, input sanitization, controlled execution policies

The comparison in Table 9 reveals a common recommendation across all papers: the importance of maintaining a strong human-in-the-loop framework. The literature consistently argues that human researchers must remain responsible for verification, interpretation, and final decision making [35, 36, 38].

9 Research Gaps

The surveyed literature shows real progress and four gaps stand out, each tied to specific evidence from the reviewed corpus.

Gap 1: No end-to-end benchmarks for research AI agents. Previous research demonstrates that isolated skills perform well independently: retrieval, summarization, tool use. Far fewer studies appraise a whole pipeline, from topic selection to literature screening, synthesis, writing support, and verification. Without that, it is near impossible to make a realistic comparison of architectures. Lack of standardized evaluation protocols is a key factor that has slowed the field down, according to Patil et al. and Akuma et al. [6, 9].

Gap 2: Factual grounding and hallucination stay unresolved in multi-step workflows. Despite reducing hallucination overall, RAG systems are not entirely free of it, particularly when retrieval performance becomes inconsistent, or the system is required to make inferences across sources that contradict each other [12, 22]. It only takes one small factual error to turn into a misleading summary or an unsupported conclusion in a research paper, which is exactly why partial grounding isn't good enough for scholarly use.

Gap 3: Long-horizon memory and cross-document reasoning remain underdeveloped. Most agentic systems cope with relatively short interaction chains without trouble, but begin to fall apart as soon as they require persistent context, cross-document consistency, or stable reasoning over a long research workflow. Memory is typically modularized as a component, and building production-ready long-term memory remains a substantial engineering challenge [21]. Cross-document consistency at the level of individual claims remains unsolved, though vector databases such as FAISS and Pinecone can help, as Sharma et al. note [25].

Gap 4: Explainability and auditability are still limited. Many systems produce fluent output but say little about why they chose a given source, how they handled conflicting evidence, or why they made a tool call. Moshkovich and Zeltyn's AgentOps framework gives observability at the operational level. It does not reach the reasoning level, where the question is *why* a decision was reasonable [29].

10 Future Directions

Future AI-augmented scientific workflows will need architectures that establish claims more rigorously, split work into sharper modules, and hold systems accountable for their own decisions.

The first priority is retrieval-grounded reasoning. A system should do more than pull documents from a database. It should show why each source is relevant, how it resolved conflicts between sources, and where the strongest evidence lies. That would build trust and reduce unsupported generation in academic settings [22, 27].

Memory, planning, and tool use also need tighter integration. Multi-step scholarly workflows need simple, reliable, context-aware tool calling, explicit task decomposition, and persistent memory of context. That points toward modular design and away from a single prompt-response paradigm [21, 28].

Multi-agent collaboration is another direction for complex research tasks. Instead of one agent performing every function, future systems may assign separate agents to retrieval, screening, synthesis, critique, and verification. Specialization can improve scalability, and a failure in one agent no longer has to take down the whole pipeline [23, 30].

Autonomous scientific discovery is the most ambitious goal, and it needs strict human oversight. Closed-loop experimentation, hypothesis generation, and robotic execution look promising. Without controls for reproducibility, provenance tracking, and safety, though, they will not produce reliable science [24].

Finally, the field needs benchmarking, cross-domain testing, and ethical alignment. Research AI agents will not count as reliable until they are evaluated on shared tasks and shared metrics, under clear governance expectations. The systems most likely to succeed will be grounded and modular, and they will combine human and AI judgment instead of relying heavily on either one.

11 Conclusion

This survey looked at how AI is used across the research process and sorted 40 papers into four categories: research guidance and lifecycle integration, literature intelligence, agentic and retrieval-augmented architectures, and operational governance (including ethics and security). The picture is mixed. AI is already used for topic exploration, retrieval, synthesis, writing assistance, and workflow orchestration. It is not yet reliable enough to work without supervision in scholarly settings, and the reviewed papers say so themselves.

The systems that succeed in these papers are hybrids. Grounding output in real sources reduces factual errors, and modular architectures let a system take on new tasks. Human oversight is still needed, because trust, reproducibility, and accountability depend on it. In the short term, the literature points toward agents that assist researchers and speed up specific parts of the research process, with transparency and human oversight kept in place throughout.

Declaration of Competing Interests. The authors declare no competing financial or non-financial interests that could have influenced the conduct or reporting of this survey.

Funding. This study was supported by Manipal Academy of Higher Education.

Data Availability. The structured coding sheet used for data extraction, the list of papers included, and thematic category assignments are available from the corresponding author upon reasonable request. All 40 included studies are cited in the reference list. No custom datasets or analytic code were generated beyond the literature corpus described in Section 2.

PRISMA 2020 Compliance Statement. This survey was conducted and reported following the PRISMA 2020 guidelines for systematic reviews [41]. Items related to meta-analysis (Items 12, 13d–f, 20b–d, 21) are not applicable as this work is a narrative systematic survey without statistical synthesis. Item 11 (risk-of-bias assessment) is addressed by the adapted, source-type-aware quality-assessment rubric in

Section 2.5 (Table 4) rather than a clinical risk-of-bias instrument, since standard tools such as the Cochrane RoB or ROBIS do not fit a corpus that mixes journal papers, conference papers, preprints, a report, and prototype/tool papers. Item 15 (certainty assessment) is addressed narratively: tool descriptions in Section 7 include explicit critical assessments that flag self-reported, non-independently-replicated performance claims wherever they appear, consistent with each source's quality-assessment rating.

12 Supplementary Material

Table 10: Comparison of AI Agent Architecture Types for Research-Oriented Systems, Organized by Taxonomic Level

Level	**Architecture Type**	**Core Components**	**Primary Strengths**	**Main Limitations**	**Scholarly Use-Case Fit**
1: Component technique	RAG [22]	Retriever, augmentation pipeline, vector databases, generation module	Grounded responses, reduced hallucination, source-backed generation	Retrieval quality strongly affects output	Literature review, citation-supported responses
	Writing Evaluation [26]	BERT fine-tuning, semantic classifiers, holistic scoring, feedback	Discipline-aware evaluation, contextual academic feedback	Training data quality, limited deep-model interpretability	Thesis-writing assistants, academic feedback
2: Agent architecture pattern	Classical Agent [20]	Reactive, deliberative, layered, learning-based; distributed coordination	Structured planning, explainable coordination, strong theoretical foundation	Limited language understanding, high deliberative overhead	Foundational reasoning, workflow coordination
	LLM-Based Agent [21]	LLM core, memory, planning, prompt engineering, tool integration	Natural-language reasoning, modular decomposition, contextual adaptation	Hallucination risk, dependency on prompt quality	Conversational research assistants, thesis guidance

Level	Architecture Type	Core Components	Primary Strengths	Main Limitations	Scholarly Use-Case Fit
	Agentic RAG [27]	Retrieval, planning, reflection, memory, tool use, multi-agent coordination	Adaptive retrieval, iterative refinement, workflow-aware reasoning	Higher latency, orchestration complexity	Autonomous research AI agents
	MAWS [23]	Distributed agents, collaborative planning, dynamic delegation, RL	Scalable coordination, resilience, parallel execution	Communication overhead, coordination failures	Multi-stage SLR: search, screening, synthesis
	Autonomous Scientific Discovery [24]	Perception, planning, experimentation, adaptation, robotic execution	Closed-loop experimentation, autonomous hypothesis generation	Explainability challenges, governance complexity	Future autonomous scientific research
	Layered Agentic NLP [25]	Perception, memory, decision engine, actuation, feedback-learning	Continuous adaptation, self-correction, multi-agent validation	Large memory infrastructure, continuous feedback tuning	Adaptive text analysis, document classification
	TacTool [28]	Reasoning LLM, tool-calling LLM, response-selection mechanism	Improved tool-use reliability, separation of reasoning and execution	Additional orchestration complexity and runtime cost	Agents using APIs, databases, retrieval tools
3: Operational management	AgentOps [29]	Observation, metrics, anomaly detection, optimization, automation	Runtime observability, debugging support, adaptive optimization	Instrumentation overhead, scalability validation gaps	Monitoring and maintaining deployed systems
4: Deployment / institutional integration	Enterprise Agentic [31]	Knowledge graphs, orchestration, perception, governance protocols	Governance-aware deployment, interoperability, hybrid autonomy	High implementation complexity, infrastructure requirements	Institution-scale research ecosystems

Level	Architecture Type	Core Components	Primary Strengths	Main Limitations	Scholarly Use-Case Fit
	Library AI Integration [32]	NLP services, analytics, recommendation, organizational adoption	Institutional integration, workflow enhancement, structured deployment	Organizational readiness and policy alignment required	Digital libraries and academic repositories

Table 11: Comparison of Existing AI-Based Research Support Tools (12 Systems)

Tool	Primary Objective	Architecture	Main Outputs	Reported Advantages	Limitations
ChatGPT-4 [13]	Thesis development support	LLM with prompt engineering, FastAPI, Python	Structured academic text, thesis scaffolding	Improves organization, reduces cognitive load	Hallucination risk, overreliance
NotebookLM	Closed-corpus QA over uploaded papers	RAG over user documents; passage-level citations	Grounded answers, source summaries	Minimal hallucination on supplied corpus	No live database retrieval
Elicit	Automated SLR assistance	LLM + Semantic Scholar retrieval; structured extraction	Ranked papers, claim extractions, comparison tables	Structured SLR-aligned outputs	Coverage limited to Semantic Scholar
Consensus	Evidence-level claim search	LLM + semantic search; evidence aggregation	Consensus meter, evidence summaries	Rapid evidence-checking	No quality weighting across studies
Semantic Scholar [11]	Academic search and discovery	Semantic indexing, NLP, citation analysis	Paper rankings, TLDR summaries, citation graphs	Large-scale coverage, TLDR useful	No native synthesis or extraction
Connected Papers [11]	Citation-graph visualization	Graph-based citation network	Visual paper network	Exploratory discovery	No extraction or summarization
Research Rabbit [11]	Interactive citation exploration	Citation graph + Zotero integration	Interactive maps, reading lists	Free, Zotero-integrated	Large seed sets hard to navigate

Tool	Primary Objective	Architecture	Main Outputs	Reported Advantages	Limitations
Litmaps [11]	Chronological research mapping	Citation timeline visualization	Temporal citation maps	Research trajectory tracking	No automation of extraction
Scite.ai	Smart citation classification	Citation classification (support/contradict/mention)	Smart citation reports	Richer signal than raw citation counts	Uneven coverage, domain variance
OmniScholar [33]	Researcher profiling, SDG mapping	CrewAI multi-agent, Semantic Scholar API, TF-IDF SDG	Profiles, SDG reports, dashboards	95%+ time reduction (self-reported)	Promotional claims; no independent eval
ScholarGenie [34]	Literature summarization, slides	CrewAI + LangChain, transformer summarizer	Summaries, slide decks, markdown	80% time reduction (self-reported)	Prompt-sensitive; promotional paper
Zotero + AI Plugins	Reference management with summaries	Plugin-dependent LLM integration	In-library summaries, research questions	Integrates with existing workflows	Plugin quality varies; not unified

A Complete Coding and Quality-Assessment Table for All 40 Included Studies

Table 12 lists every study in the final corpus (Section 2), with the coding outcome recorded on the structured sheet described in Section 2 (lifecycle stage/theme and architectural level, condensed here from the full six-field coding sheet for space) and the quality band from the rubric in Section 2.5 (Table 4). Numeric Q1–Q4 sub-scores for each paper, and the full coding sheet with all six fields, are available from the corresponding author on request; the table below reports the banded outcome (High/Moderate/Low) that those sub-scores produce, which is what the narrative sections of this survey rely on. Architectural level follows the four-level taxonomy in Section 5; entries marked "n/a (non-architectural)" are survey, policy, ethics, or empirical papers that do not themselves propose or evaluate an agent architecture and are coded only on lifecycle theme.

Table 12: Full Included-Study Disclosure: Citation, Source Type, Lifecycle Theme, Architectural Level, and Quality Band ($n = 40$)

#	Citation	Source type	Lifecycle stage / theme coded	Architectural level coded	QA band
1	[1]	Conference	Stage 1: Topic exploration & ideation	L1: profile-based recommendation	High

#	Citation	Source type	Lifecycle stage / theme coded	Architectural level coded	QA band
2	[2]	Conference	Stage 2: Literature review & knowledge extraction	L2: dual-agent reasoning pipeline	High
3	[3]	Journal	Stage 2: Literature review automation	n/a (survey)	High
4	[4]	Conference	Foundations / cross-cutting	L2: classical multi-agent patterns	Moderate
5	[5]	Conference	Foundations / cross-cutting survey	n/a (survey)	Moderate
6	[6]	Conference	Foundations / cross-cutting survey	n/a (survey)	Moderate
7	[7]	Journal	Foundations / cross-cutting survey	n/a (survey)	High
8	[8]	Preprint	Stage 2: Literature review automation	L2: multi-agent SLR system	High
9	[9]	Conference	Foundations / cross-cutting survey	n/a (survey)	Moderate
10	[10]	Preprint	Stages 3–4: end-to-end "AI Scientist"	L2: autonomous discovery (survey)	Moderate
11	[11]	Conference	Stage 2: tool ecosystem for discovery/analysis	L1: tool-ecosystem inventory	High
12	[12]	Conference (workshop)	Stage 3: scientific discovery	L2: agentic discovery patterns	High
13	[13]	Conference	Stage 4: thesis/scientific writing	L2: LLM-based agent (case study)	High
14	[14]	Preprint	Stages 2, 4, 5: comprehension, writing, peer review	n/a (survey across levels)	Moderate
15	[15]	Conference	Funding & governance	n/a (conceptual proposal)	Moderate
16	[16]	Preprint	Stages 2–3: discovery & experimentation (AI4Science)	L1/L2 (survey)	High
17	[17]	Journal	Qualitative-research practice (cross-cutting)	n/a (methodology)	Moderate
18	[18]	Institutional report	Funding & governance / institutional policy	L4: institutional integration	Moderate
19	[19]	Journal	Challenges & limitations	n/a (empirical interview study)	High
20	[20]	Conference	Foundations / cross-cutting	L1–L4: Concept-Code-Commerce framework	High
21	[21]	Conference	Foundations / cross-cutting	L2: LLM-based agent (4-component model)	High

#	Citation	Source type	Lifecycle stage / theme coded	Architectural level coded	QA band
22	[22]	Journal	Stage 2: literature review automation	L1: RAG pipeline	High
23	[23]	Preprint (SSRN)†	Stage 3 / cross-cutting workflow	L2: MAWS	Moderate
24	[24]	Conference	Stage 3: experimentation & analysis	L2: autonomous scientific discovery	High
25	[25]	Conference	Methodology / cross-cutting	L2: layered agentic NLP	High
26	[26]	Conference	Stage 4: academic writing evaluation	L1: BERT-based evaluator	High
27	[27]	Preprint	Stage 2: literature review & retrieval	L2: Agentic RAG	Moderate
28	[28]	Conference	Cross-cutting: tool-use reliability	L2: TacTool	High
29	[29]	Conference	Cross-cutting: operations & reliability	L3: AgentOps	High
30	[30]	Conference	Cross-cutting: workflow patterns	L2: agentic workflow patterns	Moderate
31	[31]	Conference	Stage 6: dissemination & governance	L4: Enterprise Agentic (IAF)	High
32	[32]	Conference	Stage 6: dissemination / library integration	L4: library AI integration	Moderate
33	[33]	Prototype/ tool	Stage 2: researcher profiling	L2: multi-agent CrewAI pipeline	Low
34	[34]	Prototype/ tool	Stages 2, 4: summarization & slide generation	L2: multi-agent CrewAI + LangChain	Low
35	[35]	Conference	Ethics & security	n/a (SLR on ethics)	Moderate
36	[36]	Journal	Ethics & security	n/a (normative/policy analysis)	High
37	[37]	Conference	Ethics & security	n/a (vulnerability analysis)	High
38	[38]	Conference (workshop)	Ethics & security	n/a (survey of security threats)	Moderate
39	[39]	Journal	Challenges & limitations / institutional	n/a (case study)	Moderate
40	[40]	Journal	Challenges & limitations	n/a (empirical ChatGPT evaluation)	High

†As noted at the corresponding bibliography entry, we could not independently confirm a formal peer-reviewed conference record for this item and cite the verifiable SSRN posting; it is scored as Moderate rather than High partly on this basis (Q2

transparency is satisfiable from the posted text, but independent verifiability of the venue is not).

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